HEAT-BLOCKING FEATURES Imaged By Ultrasound

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When power is applied, an integrated circuit begins to heat up within its package. It shortly reaches the operating temperature range for which it was designed. Excess heat begins to dissipate along a path designed for the die. If there are no structural anomalies along this path, heat dissipation from the IC will be uneventful.

For a low-power IC packaged in plastic and mounted to a printed wiring board, the dissipation path may be very simple: heat flows downward through the die attach and further dissipates through the metal die paddle. From here it makes its way through the backside mould compound. If the level of heat dissipated is low, heat-blocking anomalies along this path (die attach voids and non-bonds, for example) may have little effect, unless these cover a significant area of the dissipation path. But for higher-power dies, the absence of structural anomalies along the path is more critical and the path may terminate in a metal heat-sink designed to dissipate the higher heat flow.

The task for assemblers is to ensure that every die has a clear heat dissipation path. This means paying attention to die attach voids and surface contamination that can lead to non-bonds. For example, in plastic IC packages gaps may also occur in the bond between the backside of the die paddle and the mould compound. In insulated-gate bipolar transistor (IGBT) modules, even if no voids or other gaps are present, heat dissipation can be impaired by the tilting or warping of one or more interfaces along the dissipation path.

Die failures from imperfect heat dissipation can, to some degree, be anticipated by destructive physical analysis of field failures and of assemblies that have been life tested. These results may point to a process step that needs to be altered, or to a material that needs to be modified. A more direct method for finding structural anomalies is an acoustic micro imaging tool, which non-destructively images and analyses internal structural features, including voids, non-bonds, tilting and warping.

There are tools available that use an ultrasonic transducer to scan back and forth just above the plastic-encapsulated IC, IGBT module or other sample. The transducer is coupled to the top surface of the sample by a column of water that travels with the transducer; ultrasound at such high frequencies does not travel through the air. While moving, the transducer passes over several thousand x-y locations each second. At each location the transducer launches a pulse of ultrasound, and records the echoes sent back by material interfaces at various depths within the sample.

Ultrasound pulsed into the sample is reflected only by material interfaces. Gates are set by the operator to define the depth range from which echoes will be accepted for imaging. Put a block of homogeneous silicon onto the stage with a gate that includes virtually all of the silicon but neither the top or bottom surface, and the sample will send back no return echo signals at all, as seen at the left of Fig. 1, and the acoustic image will be solid black. If the pulse has been set to have somewhat wider gate that includes both the top and bottom surfaces (interfaces) of the silicon

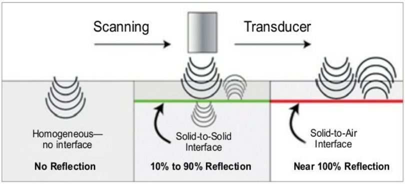

Fig. 1: Ultrasound launched into a sample is reflected not at all by a homogeneous material (left), moderately by interfaces between solids (centre) and almost completely by a solid-to-air interface (right)